BOWEI ZHAO

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Education

University of Chinese Academy of Sciences **Beijing, China**
Ph.D. in Astronomy September 2022–present
Integrated Master's–Ph.D. programme expected graduation: June 2028
Research institute: National Astronomical Observatories, Chinese Academy of Sciences (NAOC)

China Three Gorges University **Yichang, China**
B.Sc. in Optoelectronic Information Science and Engineering 2018–2022

Research Experience

Fractal Structure of Nearby Star-Forming Regions July 2023–present
Primary supervisor: Prof. Chao Liu, NAOC Secondary supervisor: Prof. Juanjuan Ren, NAOC

- Investigate the density-dependent structure of nearby star-forming regions using the 3D dust reconstruction of Leike et al. (2020).
- Independently implemented four families of fractal-dimension estimators in both 2D and 3D (box counting, perimeter–area/surface–volume, radius-of-gyration, and maximum-extent scaling).
- Found robust monotonic trends in fractal dimensions across increasing density thresholds, challenging the single-fractal picture, and revealed distinct structural variations parallel vs. perpendicular to the Galactic plane.
- Developed a Python package for density-threshold scans, structure measurements, and diagnostic plots. Conducted the project from literature review to manuscript drafting.

CSST Calibration Data System, CCDS 2022–present
Primary supervisor: Prof. Juanjuan Ren, NAOC Secondary supervisor: Prof. Chao Liu, NAOC

- Investigated calibration reference data systems used by HST and JWST, and established a local workflow for JWST CRDS best-reference-file matching.
- Define reference-file matching rules in mapping files and reference-file validation criteria in tpn files for the CSST main survey module within the CSST calibration data system (CCDS).
- Support teams working on other CSST instruments in defining calibration reference matching and validation rules.
- Led the deployment of main-survey calibration reference files to CCDS, supporting structured storage and version management for use by the scientific data-processing pipeline.
- Provide user requirements and feedback to the CCDS developer to inform improvements to the client–server system.

White Dwarf–Main-Sequence Binary Identification May 2026–present
Supervisor: Prof. Juanjuan Ren Methodological guidance: Dr. Jiadong Li

- Investigate the identification of white dwarf–main-sequence (WDMS) binary candidates using newly released SDSS and LAMOST spectroscopic data.
- Explore the application of the neural-network spectral modelling and Gaussian process classification approach of Li et al. (2025) to these datasets, focusing on data preparation and candidate-selection procedures.

Publications

[1] **Zhao, Bowei**, Wei, S., Ren, J., Wang, F., Ling, C., & Liu, C. (2025). *Calibration Reference Data Recommendation Methods*. Progress in Astronomy, 43(2), 354–374. DOI. (Review article; in Chinese with an English abstract.)

[2] Yan, Y., Wu, Y., Nie, J., et al. (including **Zhao, Bowei**) (2026). *A Robust Geometric Distortion Solution for the Main Survey Camera of CSST*. The Astronomical Journal, 171, 264. DOI.

[3] Wang, L., Chen, Z., Luo, Z., et al. (including **Zhao, Bowei**) (2026). *Comparative analysis of missing data imputation methods for CSST survey: Impact on photometric redshift estimation performance*. Astronomy & Astrophysics, 709, A239. DOI.

Manuscripts in preparation

[M1] **Zhao, Bowei**, Liu, C., Ren, J., Li, H., & Xiong, A. *On the Three-dimensional Fractal Structure of Local Star-Forming Regions*. (Manuscript in preparation.)

Technical Skills

- Programming and environment: Python; Linux shell scripting; LaTeX.
- Scientific computing: 2D/3D FITS data analysis, fractal dimension calculation, image processing, spectral analysis.
- Machine learning: developing experience in applying existing classification methods to stellar spectra.
- Astronomical data: Gaia, SDSS, LAMOST, and other astronomical surveys; 3D extinction maps; FITS image and spectral data.
- Calibration systems: CRDS/CCDS best reference-file matching, mapping rules, tpn validation criteria, and reference-file version management.

Observational Experience

Xinglong Observatory, 2.16-m Telescope

Team observing programme: *Investigating the Binary Origin of Metal-rich "Blue Horizontal-Branch" Stars*

Principal investigator: Chao Liu, NAOC

- Participated as a graduate-student observer in three observing runs within the team's BFOSC time-series spectroscopy programme.
- Assisted the telescope observing assistant with target acquisition and exposure-time decisions based on observing conditions, and prioritised nightly targets according to visibility and available observing time.
- Performed subsequent preprocessing of the acquired stellar spectra.
- Observing runs: February 22–23, 2025; May 30–31, 2024; November 9–10, 2023.

Conferences, Presentations and Schools

First National Symposium on Astrochemistry
Shenzhen, China — November 27–30, 2025
Oral presentation: *On the Three-dimensional Fractal Structure of Local Star-Forming Regions.*

ADASS XXXV: Co-authored Conference Poster
Görlitz, Germany — November 9–13, 2025
Ren, J., et al. (including **Zhao, Bowei**). *CSST On-Orbit Calibration Strategy and Data System*.
Co-author; not the presenting author.

2025 Symposium on Molecular Clouds and Star Formation
Emei, Sichuan, China — August 5–8, 2025
Participant; Southwest Jiaotong University, Emei Campus.

CSST 2025 Annual Science Meeting and Early Science Workshop
Participant; Kunming, China — July 26–30, 2025

CSST Graduate Summer School on Galaxy Science
Participant; Peking University, Beijing, China — July 17–22, 2024

CSST Science Meeting 2024
Participant; Hangzhou, China — May 13–18, 2024

CSST Science Data Processing System: 2023 Annual Review Workshop
Conference staff; Beijing, China — January 28–31, 2024

Survey Big Data Analysis Workshop I: Stellar Parameter Determination
Participant; SWIFAR, Yunnan University, Kunming, China — July 2–8, 2023

Additional Research Experience

Institute of Psychology, Chinese Academy of Sciences
Behavioral Research Assistant — March 2025–March 2026

- Administered cognitive tasks in a child brain-science study, maintaining standardized procedures and behavioral data quality.
- Interviewed parents to collect participant background and behavioral assessment data, supporting participant screening and data annotation.